

Oscillations

Key Points

Simple harmonic motion (SHM) with or without damping can have oscillations

$$\ddot{x} + k\dot{x} + \omega^2 x = 0$$

Without damping the velocity can be shown to be $v^2 = \omega^2(A^2 - x^2)$ where A is the amplitude.

For an oscillator the period $P = \frac{2\pi}{\omega}$ and the frequency $f = \frac{1}{P} = \frac{\omega}{2\pi}$

Basic Skills

Q1. State the displacement equation for SHM (without damping)

Q2. Does a vertical spring supported by the plane obey simple harmonic motion

Q3. For the damped oscillator equation $\ddot{x} + k\dot{x} + 9x = 0$

- a) State the damping constant for light damping
- b) State the damping constant for heavy damping
- c) State the damping constant for critical damping

Q4. A guitar string is vibrating at a frequency of 740Hz what is the time taken for it to complete one oscillation

Competency Skills

Q5. A speaker emits a note, vibrating at a frequency of 1800Hz with an amplitude of 2mm

- a) Show that $\omega = 3600\pi$ and hence write the differential equation for motion
- b) Find the maximum speed and maximum acceleration
- c) Find the velocity and acceleration when it is displaced by 1mm

Q6. Prove that $v^2 = \omega^2(A^2 - x^2)$ for SHM

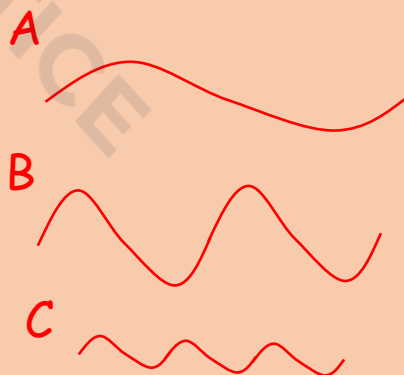
Q7. Draw a diagram showing the oscillation of SHM showing the amplitudes and how the acceleration makes the particle move back towards the centre

Q8. A woodcutting Jig-Saw does 2000 blade rotations per minute with the maximum the blade extends out of the Jig-Saw being 20cm. Find the speed of the blade when it 1cm out of the housing

Advanced Skills

Q9. To the right are 3 different waveforms for oscillators picked up during different songs in a bar

- a) Which song is the loudest
- b) Which song has the fastest tempo
- c) Is any song being dampened by the surrounding environment



Extension

Q10. Find the velocity of a particle with damped SHM

Hint: Consider the cases

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